

# MITSUBISHI GAS CHEMICAL COMPANY, INC.

Page 1 of 7

## MATERIAL SAFETY DATA SHEET

Product Name: **2-Ethylhexyl Methacrylate**

Preparation date: 8/4/09

### SECTION 1 - CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

MSDS REVISION #: 002

PRODUCT NAME: 2-Ethylhexyl Methacrylate

SYNONYMS: 2-Propenoic acid, 2-methyl-, 2-ethylhexyl ester; EHMA;  
Methacrylic acid, 2-ethylhexyl ester

DISTRIBUTED BY: Mitsubishi Gas Chemical Co., Inc.  
2-5-2 Marunouchi, Chiyoda-ku, Tokyo

PHONE NUMBERS: Inquiries - (81) 3-3283-4747

### SECTION 2 - COMPOSITION/INFORMATION ON INGREDIENTS

| <u>Component</u>                                  | <u>%</u> | <u>CAS No.</u> | <u>Exposure Limits</u>          |
|---------------------------------------------------|----------|----------------|---------------------------------|
| 2-Propenoic acid, 2-methyl,<br>2-ethylhexyl ester | >99      | 688-84-6       | Not established                 |
| Phenol, 4-methoxy-                                | <0.1     | 150-76-5       | ACGIH TLV - 5 mg/M <sup>3</sup> |

### SECTION 3 - HAZARDS IDENTIFICATION

#### \*\*\*\*\* EMERGENCY OVERVIEW \*\*\*\*\*

Material is a clear, colorless, combustible liquid, which may be absorbed through the skin in toxic amounts. Excessive breathing of vapors can cause irritation of the respiratory tract, dizziness and damage to liver and kidneys. Causes irritation to the skin, eyes and mucous membranes.

Wear impervious gloves, goggles and protective clothing when handling this product. Use only in well-ventilated area. Wash thoroughly after handling. Keep away from heat, sparks and open flames.

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Section 3 continued on next page

Product Name: **2-Ethylhexyl Methacrylate**

Preparation date: 8/4/09

### **SECTION 3 - HAZARDS IDENTIFICATION (continued)**

#### **POTENTIAL HEALTH EFFECTS**

**EYES:**

Liquid or high concentrations of vapors can cause severe irritation.

**SKIN:**

Can cause irritation, which may be severe. May cause sensitization. Persons with pre-existing skin disorders may be particularly susceptible. Methacrylates may be absorbed through the skin in amounts harmful to kidneys and liver. Persons with pre-existing liver or kidney conditions are particularly susceptible.

**INGESTION (swallowing):**

May cause irritation to mucous membranes, nausea, and CNS depression. Methacrylates may be absorbed in amounts harmful to kidneys and liver. Persons with pre-existing liver or kidney conditions are particularly susceptible.

**INHALATION (breathing):**

Can cause irritation of the mucous membranes, nausea and dizziness. High concentrations or prolonged exposure can cause damage to kidneys and liver.

**CHRONIC EFFECTS/CARCINOGENICITY:**

This product (or component) is not listed in IARC Monographs, the NTP Eleventh Annual Report or the current ACGIH TLVs as a carcinogen or potential carcinogen. OSHA does not regulate it as a carcinogen.

### **SECTION 4 - FIRST AID MEASURES**

**EYE CONTACT:**

Immediately flush with large amounts of water for at least 15 minutes, lifting upper and lower lids occasionally. Get medical attention.

**SKIN CONTACT:**

Wash exposed area with lots of soap and water. Remove contaminated clothing and launder before re-use.

Section 4 continued on next page

Product Name: **2-Ethylhexyl Methacrylate**

Preparation date: 8/4/09

**SECTION 4 - FIRST AID MEASURES (continued)****INGESTION (swallowing):**

If conscious, immediately give two large glasses of water. Never give anything by mouth to an unconscious person. Get immediate medical attention.

**INHALATION (breathing):**

If affected, move to fresh air.

**SECTION 5 - FIRE FIGHTING MEASURES****FLASH POINT:** 197° F (92° C)**FLASH POINT METHOD:** CC**UPPER EXPLOSION LIMIT:**

Unknown

**LOWER EXPLOSION LIMIT:**

Unknown

**AUTOIGNITION TEMPERATURE:**

Unknown

**EXTINGUISHING MEDIA:**

Dry chemical, carbon dioxide, water spray or alcohol foam

**FIRE AND EXPLOSION HAZARDS:**

Exposure to heat may cause polymerization resulting in a rapid release of energy, which may rupture the container explosively. Vapors are heavier than air and may travel along the ground to remote ignition sources. Vapors may then ignite explosively. All 5 gallon and larger size containers should be properly bonded and grounded.

**FIRE-FIGHTING EQUIPMENT:**

Keep personnel removed from and upwind of fire. Wear full protective clothing and self-contained breathing apparatus with full face-piece. Cool containers with water. Decomposition/combustion products may include carbon monoxide, carbon dioxide and acrid fumes.

**SECTION 6 - ACCIDENTAL RELEASE MEASURES**

Eliminate all open flames, heat, sparks and other ignition sources from the area of the spill. Use only explosion-proof equipment. Persons not wearing protective equipment should be excluded from the area of the spill until clean up has been completed. Dike area of spill to prevent spreading. Pump liquid to salvage tank or other containers. Remaining liquid may be absorbed on vermiculite or other absorbent material and shoveled into containers.

Product Name: **2-Ethylhexyl Methacrylate**

Preparation date: 8/4/09

## **SECTION 7 - HANDLING AND STORAGE**

### **HANDLING:**

Containers should be properly bonded and grounded before transferring contents to other containers. Use non-sparking tools and explosion-proof equipment when handling combustible liquids.

Avoid contact with skin, eyes and clothing. Avoid inhalation of vapors. Wash thoroughly after handling.

### **STORAGE:**

Keep in closed or covered containers away from heat, sparks and open flames when not in use. Do not store near oxidizing materials. Store in cool dry place with adequate ventilation. Material should be used within 6 months.

## **SECTION 8 - EXPOSURE CONTROLS/PERSONAL PROTECTION**

### **RESPIRATORY PROTECTION:**

A NIOSH/MSHA approved supplied air respirator is recommended if exposure exceeds level of overexposure. A canister-type respirator equipped with organic vapor cartridge may be used under certain circumstances (see 29 CFR 1910.134).

### **SKIN PROTECTION:**

Wear impervious protective gloves (contact your safety equipment supplier).

### **EYE PROTECTION:**

Chemical splash goggles in compliance with OSHA regulations and full face-shield are advised.

### **OTHER PROTECTION:**

Impervious clothing and boots are recommended.

### **ENGINEERING CONTROLS:**

Provide sufficient ventilation to maintain exposure below level of overexposure.

## MATERIAL SAFETY DATA SHEET

Product Name: **2-Ethylhexyl Methacrylate**

Preparation date: 8/4/09

**SECTION 9 - PHYSICAL AND CHEMICAL PROPERTIES**

|                          |                                         |
|--------------------------|-----------------------------------------|
| APPEARANCE:              | Clear, colorless liquid @ 77° F (25° C) |
| SPECIFIC GRAVITY:        | 0.883 @ 77° F (25° C)                   |
| VAPOR PRESSURE (mm Hg):  | Unavailable                             |
| VAPOR DENSITY (Air = 1): | Heavier than air                        |
| INITIAL BOILING POINT:   | 248° F (120° C) @ 18 mm Hg              |
| EVAPORATION RATE:        | Slower                                  |
| (Butyl Acetate = 1)      |                                         |
| SOLUBILITY in WATER:     | Soluble                                 |
| VOLATILE %:              | 100%                                    |
| pH:                      | Unavailable                             |
| pH METHOD:               | Not applicable                          |

**SECTION 10 - STABILITY AND REACTIVITY**

## STABILITY (conditions to avoid):

Stable under normal conditions. Do not expose to excessive heat.

## INCOMPATIBILITIES (materials to avoid):

Avoid contact with strong oxidizing agents, peroxides, strong alkalis and moisture.

## HAZARDOUS DECOMPOSITION PRODUCTS:

Decomposition may form carbon monoxide, carbon dioxide and acrid fumes.

## HAZARDOUS POLYMERIZATION:

Can occur if exposed to high temperatures or peroxides. This may result in containers rupturing explosively.

**SECTION 11 - TOXICOLOGICAL INFORMATION**

Toxicological information is based on information found in NIOSH Registry of Toxic Effects of Chemical Substances.

Intraperitoneal

Mouse LD<sub>50</sub> - 2614 mg/kg

Product Name: **2-Ethylhexyl Methacrylate**

Preparation date: 8/4/09

**SECTION 12 - ECOLOGICAL INFORMATION**

No data available

**SECTION 13 - DISPOSAL CONSIDERATIONS**

Incineration is the recommended disposal method for chemical wastes such as this product. Material collected on absorbent material may be deposited in a landfill in accordance with all applicable local, state and federal regulations.

This product, if disposed of, is not considered a hazardous waste under current RCRA definitions.

**SECTION 14 - TRANSPORT INFORMATION**

DOT description: Combustible liquid N.O.S. (2,4-dimethylbenzaldehyde), NA 1993, PG III.

Not regulated for domestic transportation in containers of 119 gallons capacity or less [49 CFR 173.150(f)]. Not regulated under current IMO and ICAO regulations.

**SECTION 15 - REGULATORY INFORMATION****TSCA INFORMATION:**

All components in this product are in compliance with TSCA Inventory requirements.

**CEPA:**

All components in this product are listed on the Canadian Domestic Substances List (DSL).

**EINECS:**

This product is on the European Inventory of Existing Chemical Substances under 211-708-6.

Section 15 continued on next page

# MITSUBISHI GAS CHEMICAL COMPANY, INC.

Page 7 of 7

## MATERIAL SAFETY DATA SHEET

Product Name: **2-Ethylhexyl Methacrylate**

Preparation date: 8/4/09

### SECTION 15 - REGULATORY INFORMATION (continued)

#### SARA 313 INFORMATION:

SARA requires submission of annual reports of release of toxic chemicals that appear in 40 CFR 372. This information must be included in all MSDS that are copied and distributed for this material.

Components present in this product at a level that could require reporting under the statute are: None

### SECTION 16 - OTHER INFORMATION

#### HAZARD RATING:

|            |   |              |
|------------|---|--------------|
| HEALTH     | 2 | 0 - LEAST    |
| FIRE       | 2 | 1 - SLIGHT   |
| REACTIVITY | 1 | 2 - MODERATE |
| OTHER      | - | 3 - HIGH     |
|            |   | 4 - EXTREME  |

HAZARD RATING METHOD: HMIS

#### REASON FOR REVISION:

Changed company address and phone number; added flash point data; changed transportation information; reviewed and updated

The product information contained herein is believed to be accurate as of the date of the Material Safety Data Sheet, and is provided without warranty, expressed or implied, as to the results of use of this information or the product to which it relates. Recipient assumes all responsibility for the use of this information and the use (alone or in combination with any other product), storage or disposal of the product, including any resultant personal injury or property damage.

\*\*\*\*END OF REPORT\*\*\*\*